Sahaj Bhadja

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Summary

Software engineer with 2+ years developing **autonomous-system research, full-stack applications, and tested backend services** using C, C++, Python, TypeScript, PyTorch, Linux, REST APIs, databases, Git, simulation, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed clean Python services from written requirements through implementation, testing, code review, and release, processing 5,000+ monthly workflows with failure isolation, validation, and 99%+ production reliability.
- Debugged performance and integration issues across backend services and databases for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through profiling, algorithms, and query optimization.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built testable full-stack components using React, Python, Flask, and AWS, serving 150+ users and 10,000+ daily requests while maintaining stable behavior through automated verification and debugging.
- Implemented concurrent batch processing and secure file ingestion for 1,000+ files per run, translating technical acceptance needs into tested functionality while reducing feedback turnaround time 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built large-scale autonomous-driving simulation and model-training pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through Python, PyTorch, algorithm development, dataset tooling, and reproducible performance validation.
- Designed a spatial-temporal architecture integrating road geometry, lane direction, ego state, and vehicle trajectories, testing 10,000 scenarios and improving closed-loop stability 15% while reducing off-road behavior.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered cross-platform software using complex object-oriented patterns and 20+ automated unit and integration tests across Linux, Windows, and macOS, reducing technical debt 20% and response times 30%.
- Root-caused complex distributed API and data-processing defects with AWS Glue and S3, improving pipeline efficiency 68% through Git workflows, code review, CI/CD, documentation, and collaborative Agile delivery.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Implemented eight modular services and React interfaces for real-time resource allocation, customer workflows, billing, and notifications, synchronizing two million records through REST APIs, WebSockets, and Azure SQL.
- Automated unit, integration, and end-to-end testing through reliable Jenkins, Docker, and Git-based deployment workflows, achieving 95%+ build validation while reducing release time 50% and review turnaround 25%.

Skills

Languages: C, C++, Python, Java, TypeScript, JavaScript, SQL, Shell, Bash
Full Stack: React, Next.js, Flask, FastAPI, Node.js, REST APIs, WebSockets, PostgreSQL, MySQL
Perception: PyTorch, TensorFlow, transformers, trajectory planning, dataset tooling, model evaluation
Systems: Linux, Unix, concurrency, multithreading, distributed systems, simulation, performance profiling
Process: Git, code review, issue tracking, unit testing, integration testing, debugging, CI/CD, Jenkins, Docker

Projects

CPU Scheduling Simulator Live | *C++, Simulation, Testing*

- Implemented six C++ scheduling algorithms with edge-case handling, interactive simulation, and Gantt visualization, validating 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20%.

HMIS Live | *TypeScript, REST APIs, Monitoring*

- Built a TypeScript healthcare application with role-based workflows, REST integrations, Sentry monitoring, and notifications, supporting 100+ daily appointments while reducing record retrieval time 50% and administration 30%.

CaptionCraft Live | *Python, PyTorch, Dataset Tooling*

- Developed and tested perception pipelines across four models, 10,000 images, and 100,000 captions, analyzing dataset failures and performance tradeoffs to improve measured system accuracy from 81% to 96%.